Sentinel Ridge Security

Network Topology

Written to

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What is a Network Topology

Network topology refers to the arrangement of different elements, like links, nodes, clients, and servers, that make up a computer network.

What Network Topology did we choose

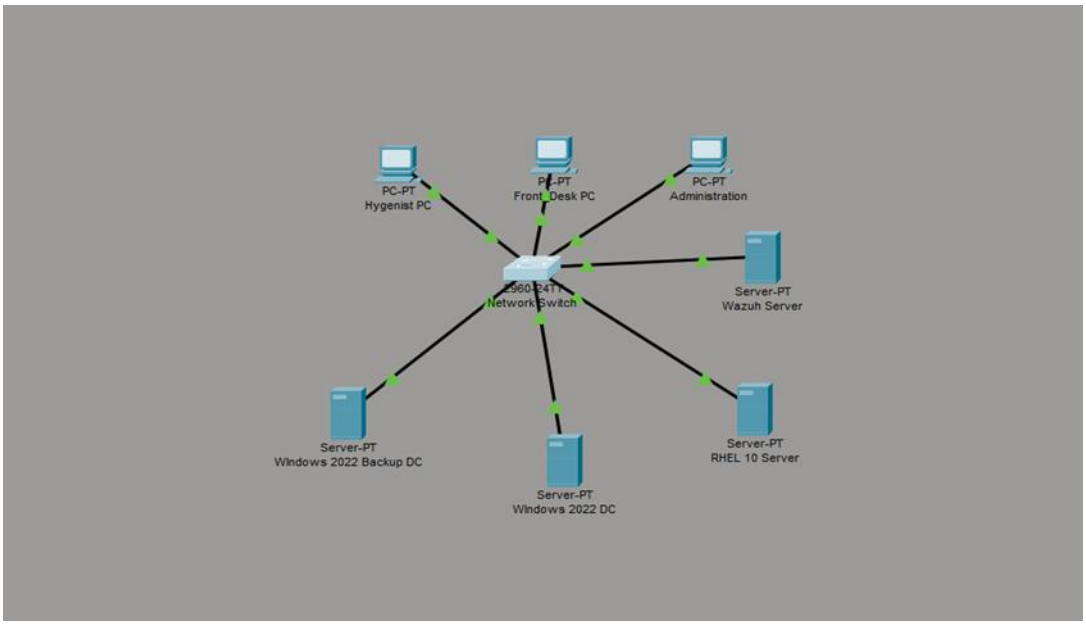
We at Sentinel Ridge security have decided to go with the Star-and-Spoke topology, or another name is the star topology. Star topology is one of the most common network layouts that is in use today. Each device is connected to a single switch. If one endpoint goes down only that one goes down. But if the central (switch) goes out, all fails.

Benefits of a Star or Hub-and-Spoke Topology

- **Simple to build and expand**: add a new PC/printer by plugging into an open switch port (or adding another access switch later).
- **Easy troubleshooting:** if one device has issues, you check **its single cable/port**; problems are localized.
- **One failure doesn't take down everyone:** if an exam room PC cable fails, only that PC goes down (not the whole office).
- **Centralized control:** one core switch is the "hub," so it's easier to manage VLANs, port security, and monitoring in one place. A network hub is not used. A network switch is used to manage traffic. "Hub" is used as a easy to understand name basis.
- **Better performance than older hub designs:** switches give dedicated bandwidth per port (less collisions, more consistent speed). Hubs (not switches) relay and can take up more bandwidth.
- **Supports segmentation cleanly:** your setup works well with **VLANs** (internal vs guest Wi-Fi) and firewall rules—important for HIPAA/PHI separation. This is what was implemented in the network diagram, guest uses WI-Fi but cannot access internal records (because of a separate VLAN). The internal infrastructure, which has access to patient data, is the only place that can access those records because the network is segmented into a different VLAN.

Tradeoff of a Star or Hub-and-Spoke Topology

The core switch is a single point of failure: if that main switch dies, most/all of the office loses connectivity.



Overcoming the trade-off

To overcome the trade-off, many organizations will keep an extra switch and contractor on standby in the case of a core outage. In the case of an outage, installing a switch and configuring it in an outage will take around two hours of in-office repair time.

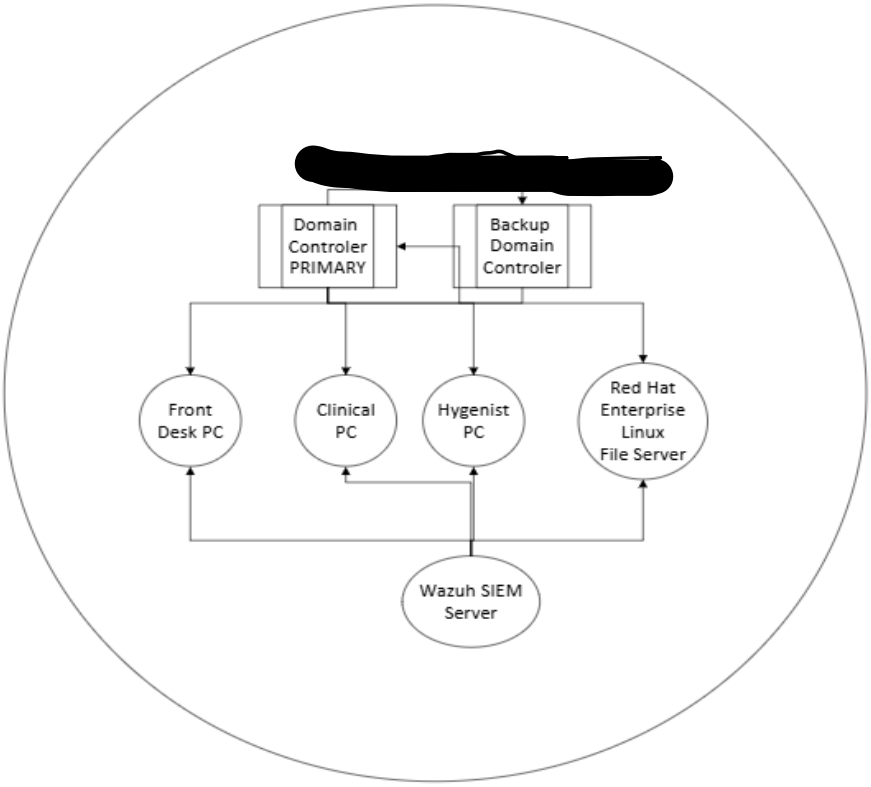
Payment: $40 per hour for an estimated 2 hours to install and configure the switch.

Domain Controller Topology

- A domain controller is a server that provides centralized authentication, authorization, and policy management for users and computers on the network.
- It plays a critical role in maintaining security, managing access, and enforcing organizational policies.

Primary Functions of the Domain Controller

- Authenticates user and computer logins
- Enforces Group Policy and other security settings
- Controls access to shared files, printers, and network resources
- Centralizes administrative control of the network environment

Why the Domain Controller Is Important

- It allows administrators to manage users, devices, and security settings from one centralized location.
- This improves consistency across systems and reduces the need to configure each workstation individually.
- It also strengthens security by ensuring that access control and policy enforcement are handled uniformly across the environment.